Cumulative Layout Shift (CLS) — Practical Proof of Concept

Internship Report — Web Performance

1. Introduction to CLS

Cumulative Layout Shift (CLS) is a web performance metric that measures how much the visible content of a webpage moves unexpectedly while the page is loading or being used. A higher CLS score means the page is less visually stable.

Unexpected movement is frustrating for users. A user may be reading a paragraph when a banner loads above it and pushes the text down, causing them to lose their place. Even worse, a button may move just before the user taps it, so they click the wrong thing by mistake. Common causes are images without set dimensions, late-loading banners or ads, and dynamically inserted content.

CLS is important because it is one of Google's Core Web Vitals. A stable page feels more professional and trustworthy, and it directly improves user experience. The standard rating thresholds are:

- **Good:** 0.10 or less
- **Needs Improvement:** above 0.10 and up to 0.25
- **Poor:** above 0.25

2. Objective

The objective of this practical is to compare two implementations of the same page and show how CLS changes between them:

- **Bad CLS:** a banner loads without reserved space and pushes the existing content downward.
- **Fixed CLS:** space is reserved for the banner before it loads, so the surrounding content does not move.

The demonstration application is built with **Angular** and uses Google's `web-vitals` library to measure CLS and display the score on the page.

3. Practical Demonstration — Before the Banner Loads

Figures 1 and 2 show both demo pages before the banner is loaded. Both pages start with a status of “Waiting to run” and a Measured CLS of 0.0000. The difference is in how the page is laid out.

In the Bad CLS page, no space is kept for the banner. The article card sits at the top of the content area, so when the banner arrives later, it can only be placed above the article and pushes it down.

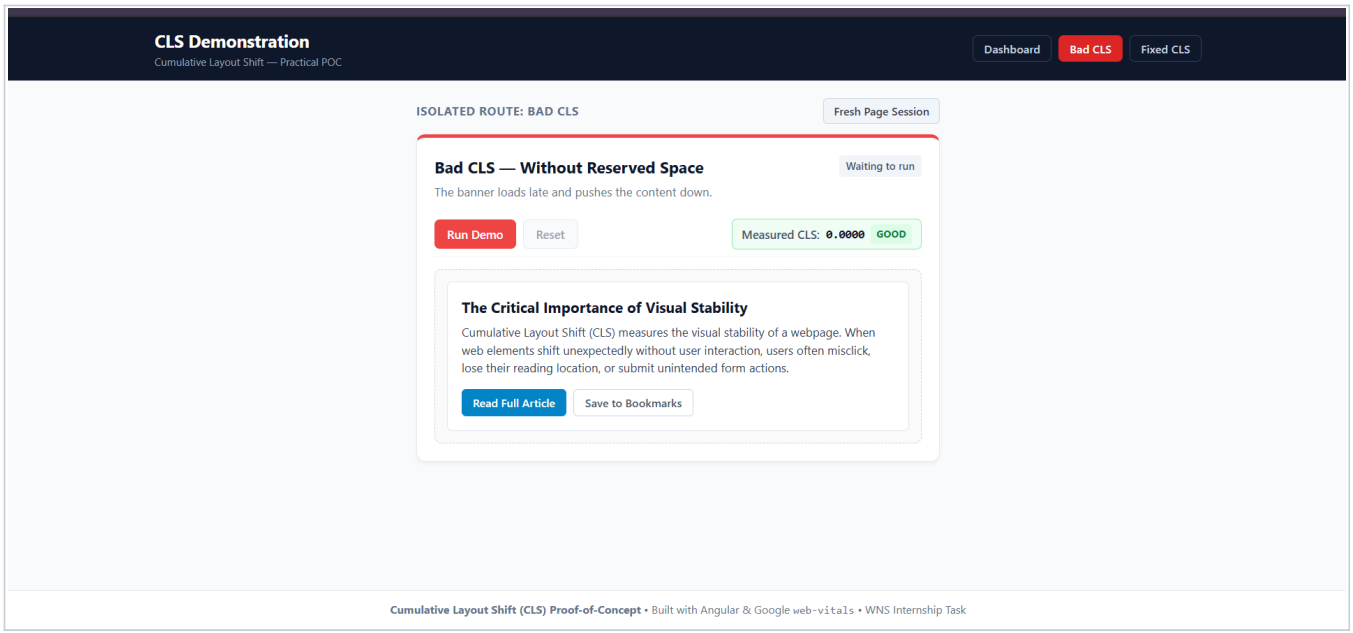


Figure 1: Bad CLS page before the banner loads.

In the Fixed CLS page, a placeholder labelled “Reserved Banner Space (220px)” holds the banner’s position in advance. The article is already placed below this space, so it has no reason to move when the banner appears.

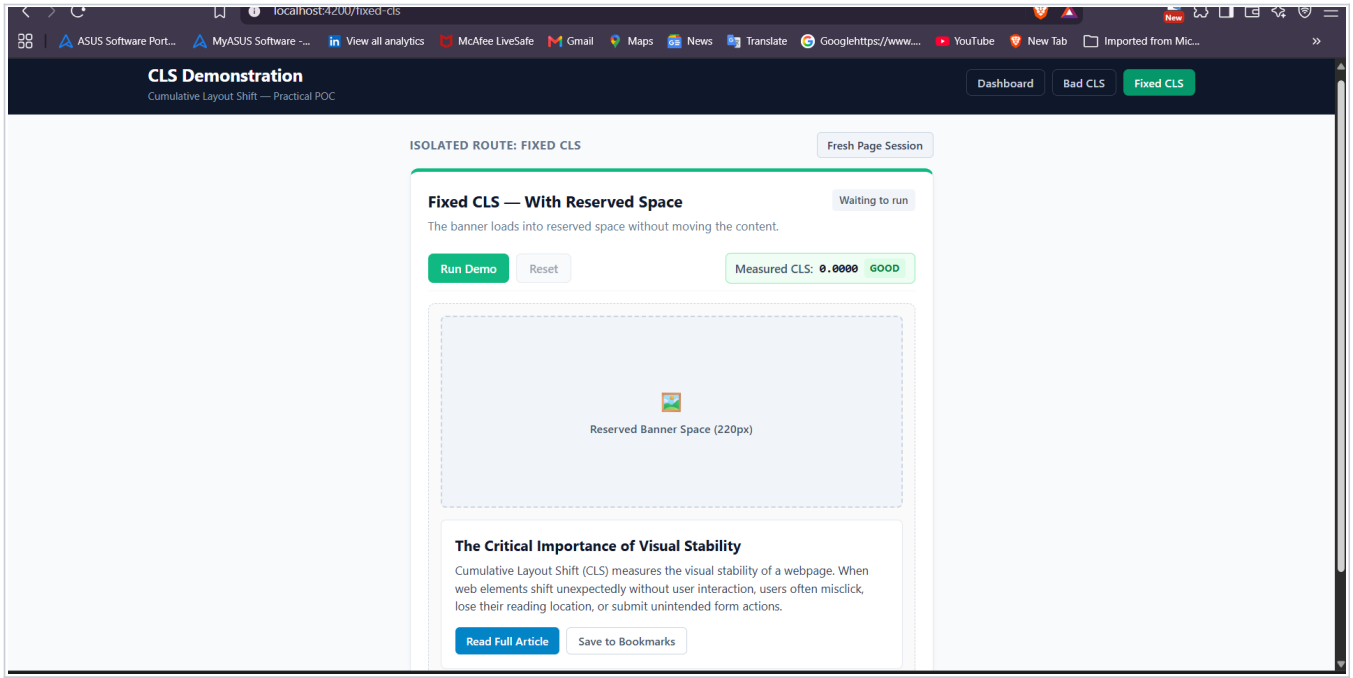


Figure 2: Fixed CLS page before the banner loads.

4. Practical Demonstration — Chrome DevTools Evidence

After clicking Run Demo, the banner loads on each page. Figures 3 and 4 show the result with the Chrome DevTools Performance panel open and a recording of the page. The Layout shifts track in DevTools marks layout-shift events, and the application shows the CLS score measured by the web-vitals library.

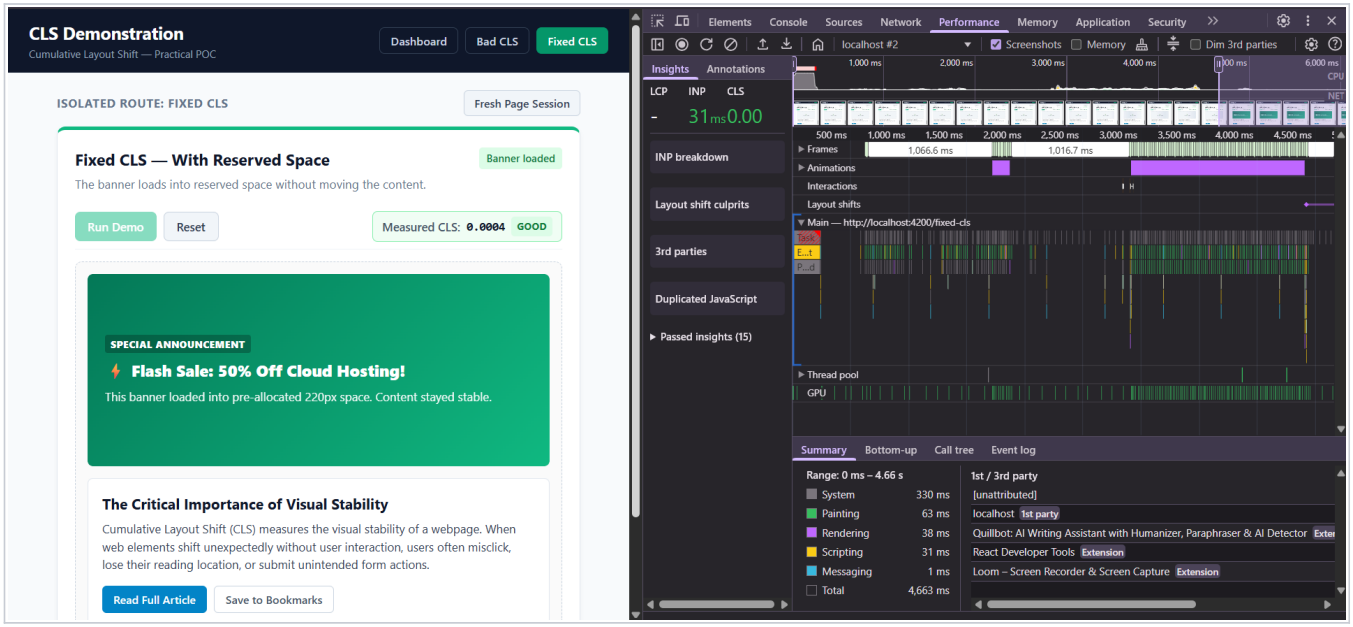


Figure 3: Fixed CLS demonstration with the banner loaded, with Chrome DevTools Performance recording.

Fixed CLS result. The banner loaded into the reserved 220px space without pushing the article downward. The application shows the status “Banner loaded” and a measured CLS of **0.0004**, rated **Good**. DevTools shows a CLS of 0.00 (rounded). This is a very small value, far below the 0.10 limit, rather than a perfect zero.

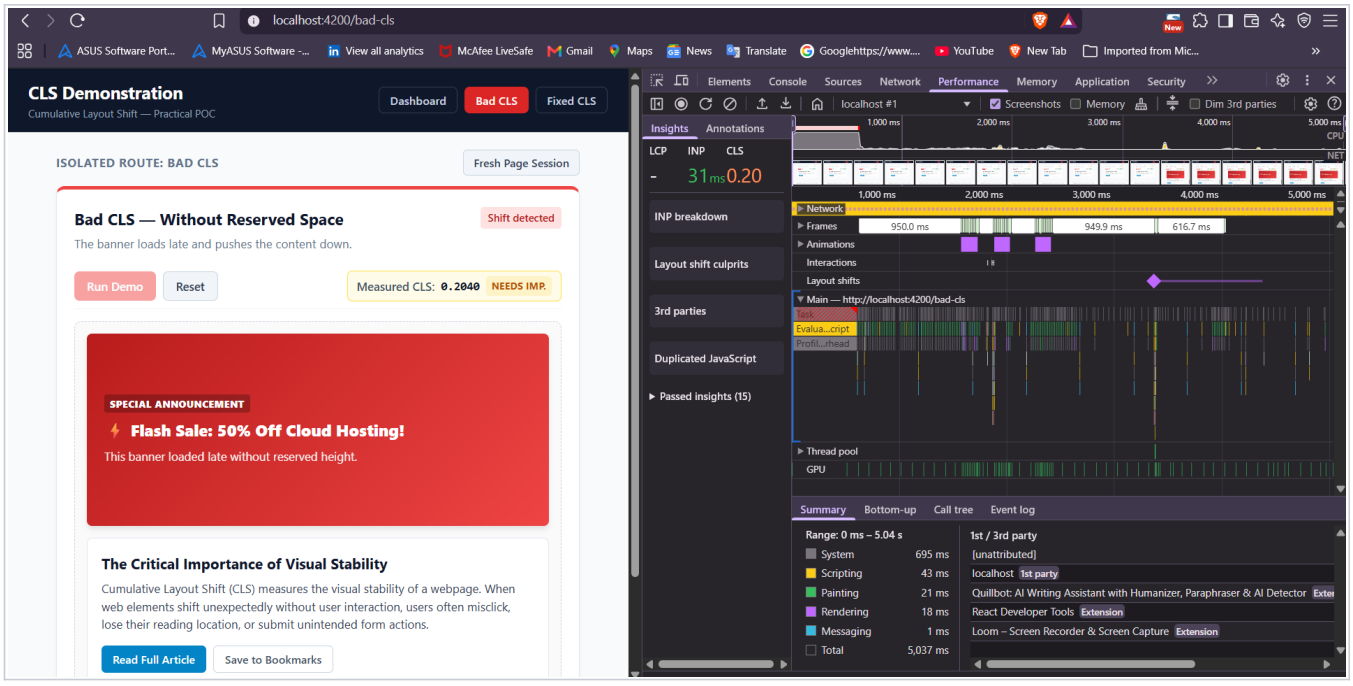


Figure 4: Bad CLS demonstration with the banner loaded, with Chrome DevTools Performance recording.

Bad CLS result. The banner appeared without reserved space and pushed the article downward. The DevTools recording shows a layout-shift event in the Layout shifts track, and DevTools reports a CLS of 0.20 (rounded). The application shows the status “Shift detected” and a measured CLS of **0.2040**, rated **Needs Improvement**.

Both pages show the same kind of banner and article content. The key difference is whether space was reserved, and that change reduced the measured CLS from 0.2040 to 0.0004. These values are the results displayed in the screenshots above.

5. Conclusion

- Content that is inserted dynamically, such as a banner, can cause unexpected layout shifts if space is not reserved for it.
- Reserving space for banners and images in advance (a 220px placeholder in this demo) keeps the page visually stable and moved the score from Needs Improvement to Good.
- Chrome DevTools Performance and the `web-vitals` library help developers investigate and measure CLS, so layout problems can be found and fixed early.